SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
SAVEETHA SCHOOL OF ENGINEERING

ITA0606 – MACHINE LEARNING

ASSIGNMENT 1

DESIGN, ANALYSE AND VALIDATE AN INTELLIGENT
EARLY-WARNING DECISION SUPPORT SYSTEM
FOR ACADEMIC RISK IDENTIFICATION
USING CONCEPT LEARNING AND DECISION TREE LEARNING

SUBMITTED BY

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DECLARATION

I, T. M. Rishika (192412200), declare that this assignment report has been prepared by me for the course ITA06 – Machine Learning. The explanations, calculations, implementation design and analysis presented in this report have been developed for the stated academic problem. All external concepts and resources used for learning are acknowledged in the references.

I understand that academic integrity requires honest representation of work, transparent reporting of experimental results and appropriate acknowledgement of sources.

ACKNOWLEDGEMENT

I express my sincere gratitude to the faculty members of Saveetha School of Engineering for providing the opportunity to study machine-learning concepts through an application-oriented assignment. The university academic early-warning scenario helped connect theoretical ideas such as concept learning, version spaces, entropy and heuristic search with a practical decision-support problem.

I also acknowledge the authors and maintainers of the textbooks, documentation and open educational resources consulted while preparing the theoretical explanations and implementation plan. The supplied assignment datasets and task descriptions form the basis of the calculations in this report.

ABSTRACT

This report investigates an interpretable early-warning system intended to identify university students who may be at risk of failing. The study compares two supervised learning approaches: conjunctive concept learning with Candidate Elimination and decision-tree learning using ID3. The supplied five-example concept-learning dataset is used to study hypothesis representation, version spaces and inductive bias. The fourteen-example decision-tree dataset and six held-out records are used to calculate entropy, information gain, tree structure and generalisation behaviour.

A reproducible synthetic dataset containing 500 student records is generated for scalability experiments. Its categorical attributes follow the definitions in the assignment, and its target labels are produced by a documented risk-score rule with stochastic variation. The report separates results computed directly from the supplied datasets from illustrative synthetic experiments. It examines label noise, unseen categories, recall-oriented evaluation, pruning and human-readable explanations.

The central conclusion is that a small, exact conjunctive learner is valuable for demonstrating version-space theory but is too restrictive for the changing deployment conditions. A controlled decision-tree baseline is interpretable and practical, but should be supported by class-sensitive evaluation, careful validation, explicit handling of unknown values and human review. A redesigned pipeline is therefore recommended for the Day-4 challenge.

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PROBLEM SCENARIO AND DATASET DEFINITIONS

The assignment describes a university that wants to identify students who may fail early enough to offer counselling, remedial teaching or mentoring. Records are collected after Week 6. The prediction target, AtRisk, is assigned by the academic council at the end of the semester. This timing creates a supervised learning task: the historical target is known during training, while the model's output for a current student is an estimate that must be reviewed before intervention.

Table 1. Supporting data and analysis

Attribute	Allowed values	Operational definition	Role
Attendance	Low / Medium / High	Low <65%; Medium 65–80%; High >80%	Engagement proxy
Assignments	Irregular / Regular	Regular means at least 80% submitted on time	Task completion
StudyHours	Low / High	Low means fewer than 8 self-reported hours/week	Study effort
PriorCGPA	Weak / Strong	Weak <6.0; Strong ≥6.0	Prior attainment
LMS	Passive / Active	Activity on the learning-management system	Digital engagement
Mentoring	No / Yes	Enrolled in mentoring programme	Support exposure
AtRisk	Yes / No	Council-assigned end-semester label	Target

Dataset A contains five examples and six predictors. Dataset B contains fourteen examples and four predictors. Test Set T contains six held-out records. Dataset C must contain at least 500 examples; this report generates exactly 500 records using a fixed random seed and a disclosed synthetic rule. The synthetic dataset is not presented as real university data.

TASK 1 – IDENTIFY AND FORMULATE THE ML PROBLEM

1.1 Formal learning problem: Task, performance and experience

Task (T): Given the categorical profile of a student observed after Week 6, assign a binary early-warning estimate, AtRisk = Yes or AtRisk = No. The model is not deciding whether a student deserves support; it is prioritising records for a human advisor to review.

Performance measure (P): The primary deployment measure is recall for the AtRisk = Yes class, because a false negative can leave a student without timely assistance. Precision, F1-score, specificity, balanced accuracy and the confusion matrix must also be reported. Accuracy alone is insufficient when the cost of missing an at-risk student is high.

Experience (E): The learner receives labelled historical records containing the predictors and the council-assigned AtRisk outcome. The small labelled sets A and B are used for hand calculations; the held-out set T is reserved for evaluation. Dataset C is used only for scalable synthetic experiments.

1.2 Attribute relevance and risk

Table 2. Supporting data and analysis

Attribute	Type	Values	Plausible relevance	Risk / limitation
Attendance	Categorical ordinal	Low, Medium, High	Absence can reduce access to lectures and labs	May reflect health, travel or timetable barriers
Assignments	Categorical	Irregular, Regular	Missed submissions can indicate difficulty or time pressure	Late work may result from access or disability issues
StudyHours	Categorical ordinal	Low, High	Study time can reflect preparation effort	Self-report is noisy and does not measure quality
PriorCGPA	Categorical ordinal	Weak, Strong	Prior attainment may capture accumulated academic difficulty	Can reproduce historical disadvantage
LMS	Categorical	Passive, Active	Online participation can indicate course engagement	Offline study may be substantial
Mentoring	Categorical	No, Yes	Support participation may reveal intervention context	Potentially misleading due to selection effects

Mentoring is a particularly risky predictor. A student may be enrolled because an advisor already suspects academic difficulty. The attribute can therefore encode a previous human decision rather than an independent cause of risk. If the model learns that mentoring implies risk, it may label a student as high risk because support was offered. A safer initial design either excludes Mentoring from the prediction model or evaluates it separately with a clear causal and fairness review.

1.3 Boolean target concept and instance-space size

Let x = (a, s, h, c, l, m) denote the six categorical predictor values in Dataset A. The target concept is c(x) = 1 when AtRisk = Yes and c(x) = 0 when AtRisk = No. The instance space is the Cartesian product of the six attribute domains.

For Dataset A, $|X_A| = 3 \times 2 \times 2 \times 2 \times 2 \times 2 = 96$ possible predictor combinations. Five examples cover $5/96 \times 100 = 5.2083\%$ of the space.

For Dataset B, the four predictor cardinalities are 3, 2, 2 and 2. Therefore $|X_B| = 3 \times 2 \times 2 \times 2 = 24$. The fourteen training records cover $14/24 \times 100 = 58.3333\%$ of the space. This is a much larger fraction than Dataset A, but still does not guarantee coverage of every combination.

Table 3. Supporting data and analysis

Dataset	Predictors	Instance-space calculation	Space size	Observed records	Coverage
A	6	3×2×2×2×2×2	96	5	5.2083%
B	4	3×2×2×2	24	14	58.3333%
T	4	Same as B	24	6 held-out	Not training coverage

Dataset	Predictors	Instance-space calculation	Space size	Observed records	Coverage
C	4	Same as B	24	500 generated	Repeated observations

1.4 Learning paradigm

The core task is supervised binary classification because each historical record contains a known target label assigned by the academic council. The model learns a mapping from observed attributes to the target. Unsupervised clustering may help explore behavioural profiles, while reinforcement learning may later help select interventions, but neither replaces the initial labelled risk classifier.

1.5 Application-specific machine-learning issues

Table 4. Supporting data and analysis

Issue	Evidence in this scenario	Potential effect	Mitigation
Small sample	Only 5 examples in A and 14 in B	Unstable rules and high variance	Collect more semesters; cross-validate
Label noise	Different advisors may disagree	Contradictory training targets	Audit labels; adjudication; noise-aware training
Sampling bias	Historical records may omit some student groups	Unequal error rates	Stratified evaluation and subgroup audits
Measurement error	StudyHours is self-reported	Incorrect splits	Use validated logs where appropriate; mark missingness
Concept drift	New online study modes alter attendance patterns	Old rules become stale	Monitor distributions and retrain after review
Privacy	Academic records are sensitive	Unauthorised exposure	Data minimisation, access controls and retention limits
Proxy discrimination	PriorCGPA or Mentoring may proxy disadvantage	Unfair flags	Fairness review and human appeal process
Class imbalance	Future cohorts may contain fewer at-risk students	Accuracy can hide missed positives	Recall, PR curve and class-sensitive thresholds

1.6 Error-cost argument

A false negative means that a student who is actually at risk is predicted as safe. The student may miss an opportunity for timely counselling or remedial support. A false positive consumes advisor time and may cause unnecessary concern, but it can usually be resolved through a short review. For this reason, the system should emphasise recall while monitoring precision so that the support team is not overwhelmed.

The prediction must be framed as a screening signal, not a final judgement. The student should be able to ask for correction of inaccurate information, and an advisor should be able to override the model with a recorded reason.

TASK 2 – ANALYSE THE LEARNING PARADIGM

2.1 Comparison in the early-warning context

Table 5. Supporting data and analysis

Criterion	Supervised learning	Unsupervised learning	Reinforcement learning
Labels	Council outcome labels are available	No outcome labels required	Reward arrives after actions
Output	AtRisk probability/class	Clusters or latent profiles	Policy for selecting actions
Feedback	Known target during training	Structure and stability of groups	Reward or penalty over time
Data needed	Student predictors with outcomes	Student predictors only	States, actions and outcome feedback
Success evaluation	Recall, precision, F1, calibration	Cluster stability and usefulness	Cumulative reward and policy safety
Advisor interpretability	Rules/tree paths can be shown	Profiles need interpretation	Policy may be difficult to explain
Main risk	Label bias and overfitting	Clusters may not align with risk	Unsafe exploration and delayed reward

2.2 A concrete unsupervised use

Clustering could group students by engagement patterns, for example low attendance with active LMS use, or high attendance with irregular assignment completion. Advisors could use these profiles to design different support programmes. However, a cluster label is not the same as AtRisk. Clustering cannot, by itself, determine which student will fail because no outcome target is used to define the groups.

2.3 A plausible reinforcement-learning follow-up

A later intervention-planning problem can be expressed with state s = (current risk estimate, attendance trend, assignment trend, prior interventions), action a = (offer counselling, assign mentor, recommend remedial class, or no additional action), and reward r based on subsequent academic improvement, student well-being and intervention burden. The reward must be designed carefully to avoid penalising students for circumstances outside their control.

Reinforcement learning is unsuitable for the initial risk-classification task because the assignment supplies historical labelled examples, not sequential action-and-reward trajectories. Exploration by trying different interventions on students would also raise ethical and operational concerns.

2.4 Selection, counterargument and hybrid

Supervised learning is the appropriate paradigm for the core classifier. The strongest objection is that the labels are assigned at the end of the semester and may contain advisor bias. This is a real limitation, but it does not make unsupervised learning a substitute for the requested target. The response is to audit labels, measure subgroup errors and preserve human review.

A useful hybrid is supervised classification plus unsupervised profile analysis. The classifier estimates risk, while clustering reveals behavioural subgroups that may need different interventions. The two outputs should remain distinct: risk prediction must be validated against labels, and clusters must not be treated as ground truth.

TASK 3 – FORMULATE THE CONCEPT-LEARNING TASK

3.1 Target concept and examples

For Dataset A, define c: X → {0,1}, where c(x)=1 represents AtRisk = Yes and c(x)=0 represents AtRisk = No. Positive examples are E1, E2 and E4. Negative examples are E3 and E5.

Table 6. Supporting data and analysis

Example	Attendance	Assignments	StudyHours	PriorCGPA	LMS	Mentoring	Label
E1	Low	Irregular	Low	Weak	Passive	No	Yes
E2	Low	Irregular	Low	Strong	Passive	No	Yes
E3	High	Regular	High	Strong	Active	Yes	No
E4	Low	Irregular	High	Weak	Passive	Yes	Yes
E5	Medium	Irregular	High	Strong	Active	No	No

3.2 Hypothesis representation

A hypothesis is a six-position conjunction. Each position contains a specific domain value, '?' for any value, or '∅' for no value. A tuple covers an instance when every specific constraint matches the corresponding attribute. A hypothesis containing at least one empty constraint covers no instances.

The most general hypothesis is ⟨?,?,?,?,?,?⟩. The most specific hypothesis is ⟨∅,∅,∅,∅,∅,∅⟩. E1 as a singleton hypothesis is ⟨Low, Irregular, Low, Weak, Passive, No⟩. E3 is ⟨High, Regular, High, Strong, Active, Yes⟩.

3.3 Hypothesis-space counts

For an attribute with d values, its syntactic position can contain one of d values, '?', or '∅', giving d+2 choices. The syntactic count is therefore (3+2)(2+2)^5 = 5×4^5 = 5,120.

If empty constraints are excluded, each position has d+1 non-empty choices. The number of non-empty conjunctive concepts is (3+1)(2+1)^5 = 4×3^5 = 972. Adding the single empty concept gives 973 semantically distinct hypotheses. The instance space contains 96 points, as calculated in Task 1.

Table 7. Supporting data and analysis

Quantity	Calculation	Result
Syntactically distinct representations	(3+2)(2+2)^5	5,120
Semantically distinct conjunctions	(3+1)(2+1)^5 + empty concept	973
Instance-space points	3×2×2×2×2×2	96

3.4 Generality ordering example

Choose h1 = ⟨?, ?, ?, ?, ?, ?⟩ and h2 = ⟨Low, ?, ?, ?, ?, ?⟩. Every instance covered by h2 is also covered by h1, so h1 is more general than h2. Let h3 = ⟨?, Regular, ?, ?, ?, ?⟩. It is incomparable with h2: neither hypothesis subsumes the other because they constrain different attributes.

3.5 Consistency checks

Table 8. Supporting data and analysis

Hypothesis	Constraint tuple	Result	Reason
h_a	⟨Low, ?, ?, ?, Passive, ?⟩	Consistent	Covers E1, E2 and E4, all positive; no negative example matches.
h_b	⟨?, Irregular, ?, ?, ?, ?⟩	Inconsistent	Covers E5, which is negative.
h_c	⟨Low, Irregular, Low, ?, ?, ?⟩	Consistent	Covers E1 and E2; both are positive.

Satisfying an instance means that the instance matches the constraints of one hypothesis. Consistency with training data is stronger: every covered positive example must be labelled positive, and every covered negative example must be excluded. A hypothesis may cover a new instance without being consistent with the complete training set.

TASK 4 – APPLY THE CANDIDATE ELIMINATION ALGORITHM

4.1 Initial boundaries and update logic

Candidate Elimination maintains a specific boundary S and a general boundary G for the set of hypotheses consistent with the observations. Initially S is the most specific hypothesis and G contains the most general hypothesis. On a positive example, S is minimally generalised to cover the example and any member of G that fails to cover it is removed. On a negative example, each member of G that covers the negative example is minimally specialised, subject to remaining at least as general as S.

The algorithm is exact only under its representation and label assumptions. In this assignment, examples are processed in the supplied order E1 through E5. Because the dataset is very small, the boundary trace is more informative than a final boundary alone.

4.2 Step-by-step trace

The following trace uses the standard conjunctive representation. A specific value is written explicitly, '?' denotes an unconstrained attribute, and ∅ denotes an impossible constraint. Boundary members are shown as sets; equivalent duplicate members are removed.

Table 9. Supporting data and analysis

Step	Example	Label	Action	S after step	G after step	Reason
0	—	—	Initialise	⟨∅,∅,∅,∅,∅,∅⟩	{⟨?,?,?,?,?,?⟩}	No observations processed.
1	E1	Positive	Generalise S to E1	⟨Low,Irregular,Low,Weak,Passive,No⟩	{⟨?,?,?,?,?,?⟩}	First positive example fixes S to the observed tuple.
2	E2	Positive	Minimally generalise differing PriorCGPA	⟨Low,Irregular,Low,?,Passive,No⟩	{⟨?,?,?,?,?,?⟩}	E2 differs from E1 only in PriorCGPA.
3	E3	Negative	Specialise G to exclude E3 while remaining ≥ S	S unchanged	Boundary specialisations retained only if they cover all S-positive instances.	E3 is negative; G must exclude it.
4	E4	Positive	Generalise S for StudyHours and Mentoring	⟨Low,Irregular,?, ?,Passive,?⟩	Remove G members not covering E4	E4 is positive and differs from S on StudyHours and Mentoring.
5	E5	Negative	Specialise remaining G members that cover E5	S unchanged	Remove any G member covering E5; retain only minimal valid specialisations.	E5 is negative.

Important implementation note: Candidate Elimination's exact G boundary requires enumerating all minimal specialisations and removing dominated hypotheses. The implementation in Appendix A performs this operation over the finite attribute domains and prints the resulting boundaries. This report does not replace the trace with an unsupported hand-written final set; the executable routine is the reproducibility reference for the boundary members.

4.3 Positive and negative update reasoning

For a positive observation, each position in S is compared with the corresponding example value. A mismatch between two specific values is replaced by '?'; an already general position remains '?'. The operation is minimal because no narrower constraint could cover both positive examples.

For a negative observation, a general hypothesis that already excludes the negative example remains unchanged. A hypothesis that covers the negative example must be specialised on one or more attributes. A specialisation is invalid if it excludes a previously observed positive example, or if it is more specific than S. If one retained specialisation is more general than another, the more specific duplicate is removed from G.

Exact boundary verification by finite enumeration

To remove ambiguity in the hand trace, the finite hypothesis language was enumerated after each prefix of Dataset A. For every prefix, all hypotheses were tested against every processed label. S is the set of maximally specific consistent hypotheses; G is the set of maximally general consistent hypotheses. This direct enumeration is a check on the boundary definitions and gives the following exact counts.

Table 10. Supporting data and analysis

Processed examples	|S|	S boundary	G boundary	Version-space size
E1	1	⟨Low, Irregular, Low, Weak, Passive, No⟩	⟨?,?,?,?,?,?⟩	64
E1–E2	1	⟨Low, Irregular, Low, ?, Passive, No⟩	⟨?,?,?,?,?,?⟩	32
E1–E3	5	⟨Low, Irregular, Low, ?, Passive, No⟩	⟨Low,?,?,?,?,?⟩; ⟨?,Irregular,?,?,?,?⟩; ⟨?,?,Low,?,?,?⟩; ⟨?,?,?,?,Passive,?⟩; ⟨?,?,?,?,?,No⟩	31
E1–E4	3	⟨Low, Irregular, ?, ?, Passive, ?⟩	⟨Low,?,?,?,?,?⟩; ⟨?,Irregular,?,?,?,?⟩; ⟨?,?,?,?,Passive,?⟩	7
E1–E5	2	⟨Low,?,?,?,?,?⟩; ⟨?,?,?,?,Passive,?⟩	⟨Low,Irregular,?,?,Passive,?⟩	6

Interpretation: after the negative example E3, the general boundary is specialised to exclude E3 while still covering E1 and E2. The positive E4 then forces the specific boundary to generalise over StudyHours and Mentoring. E5 removes hypotheses that would incorrectly cover that negative example. The final version space contains six consistent conjunctions.

4.4 Boundary interpretation

The version space is the set of all hypotheses between S and G that remain consistent with every processed example. A boundary can become empty if the data contain contradictions that cannot be represented by the chosen hypothesis language. This is not a software crash condition: it is a diagnostic that the representation, labels or data require review.

TASK 5 – ANALYSE VERSION SPACE AND INDUCTIVE BIAS

5.1 Version space

The version space is the collection of all hypotheses in the selected hypothesis language that classify every observed positive and negative example correctly. It is not the entire hypothesis space: it is the subset that survives the evidence. In a boundary representation, a hypothesis belongs to the version space when it is at least as general as some member of S and no more general than some member of G.

The exact enumeration is generated by the Candidate Elimination code in Appendix A. The enumeration routine tests each legal conjunctive hypothesis against E1–E5 and retains only consistent hypotheses. This direct check is also used as an independent validation of the boundary representation.

The six consistent hypotheses are:

v1 = ⟨Low, Irregular, ?, ?, Passive, ?⟩

v2 = ⟨Low, Irregular, ?, ?, ?, ?⟩

v3 = ⟨Low, ?, ?, ?, Passive, ?⟩

v4 = ⟨Low, ?, ?, ?, ?, ?⟩

v5 = ⟨?, Irregular, ?, ?, Passive, ?⟩

v6 = ⟨?, ?, ?, ?, Passive, ?⟩

5.2 Query voting and ambiguity

Table 11. Supporting data and analysis

Query	Instance	Interpretation
Q1	⟨Low, Regular, High, Weak, Passive, No⟩	The query combines a low-attendance value with regular assignments and other mixed attributes; hypotheses that constrain only one of these dimensions may disagree.
Q2	⟨Low, Irregular, Low, Strong, Passive, Yes⟩	This query closely resembles E1/E2 and is expected to receive stronger agreement from hypotheses consistent with the positive examples.

For each query, every version-space hypothesis votes Yes if it covers the query and No otherwise. The vote fraction is a useful measure of ambiguity, but it is not a calibrated probability. A query with votes split near 50:50 is a strong candidate for the next label because its outcome is likely to eliminate a large portion of the current version space.

The most informative next example is not selected by intuition alone. The implementation can score every possible instance by how evenly the current version space divides its votes, then request an expert label for the highest-scoring instance. In practice, the advisor should also consider whether the case is ethically appropriate and whether the label can be obtained reliably.

Exact query-vote check for the six final hypotheses

Table 12. Supporting data and analysis

Query	Yes votes	No votes	Vote split	Interpretation
Q1	3	3	50% / 50%	Ambiguous; useful for requesting a new label.
Q2	6	0	100% / 0%	All six hypotheses classify the instance as positive.

The six final hypotheses were enumerated from Dataset A. Q1 receives an even split, while Q2 is covered by all six. The Q1 label is therefore more informative for reducing version-space uncertainty, assuming an expert can provide a reliable label.

5.3 Four related concepts

Table 13. Supporting data and analysis

Concept	Meaning	Example in this assignment
Training data	Observed labelled examples	E1–E5 in Dataset A
Hypothesis space	All allowed candidate rules	All 973 semantically distinct conjunctions
Version space	Hypotheses consistent with observed examples	Candidates that cover all positives and exclude all negatives
Inductive bias	Assumptions restricting what can be learned	Only conjunctions of single values and wildcards are considered

5.4 Inductive bias and added example

Candidate Elimination assumes the target can be represented by the chosen conjunctive language and that labels are consistent. Add the positive example ⟨High, Regular, High, Strong, Passive, No⟩ after E1–E5. This example conflicts with any candidate that requires Low attendance, Irregular assignments, or Passive LMS in a way that excludes it. The version space must be recomputed; it may shrink substantially or become empty if the positive example cannot be covered without also covering a previously observed negative example.

If the hypothesis space were the power set of the 96-point instance space, the number of possible Boolean concepts would be 2^96 = 79,228,162,514,264,337,593,543,950,336. Such a space can represent every possible labelling, but without a preference for simpler or structured concepts it provides no useful generalisation to unseen points. Inductive bias is therefore necessary for learning from finite observations.

TASK 6 – CONSTRUCT THE DECISION TREE

6.1 Entropy of Dataset B

Dataset B contains 14 records: 7 AtRisk=Yes and 7 AtRisk=No. The class proportions are p(Yes)=0.5000 and p(No)=0.5000. Entropy $H(S)=-\Sigma p_i \log_2 p_i = 1.0000$ bits. The balanced class distribution gives the maximum binary entropy of 1 bit.

6.2 Root information gain

Table 14. Supporting data and analysis

Attribute	Branch value	n	Yes	No	Entropy / gain
Attendance	Low	5	4	1	0.7219
Attendance	Medium	5	2	3	0.9710
Attendance	High	4	1	3	0.8113
Attendance	Weighted H / Gain				IG=0.1636
Assignments	Irregular	8	5	3	0.9544
Assignments	Regular	6	2	4	0.9183
Assignments	Weighted H / Gain				IG=0.0611
LMS	Passive	7	5	2	0.8631
LMS	Active	7	2	5	0.8631
LMS	Weighted H / Gain				IG=0.1369
PriorCGPA	Weak	7	6	1	0.5917
PriorCGPA	Strong	7	1	6	0.5917
PriorCGPA	Weighted H / Gain				IG=0.4083

PriorCGPA has the largest root information gain (0.4083 bits), so it is selected as the root. This split creates two branches with seven examples each. Each branch is still impure, so ID3 continues recursively.

6.3 Recursive split calculations

PriorCGPA = Weak: 7 examples; class counts {'Yes': 6, 'No': 1}; entropy = 0.5917 bits.

Attendance: information gain = 0.3060 bits.

Assignments: information gain = 0.1981 bits.

LMS: information gain = 0.1981 bits.

PriorCGPA = Strong: 7 examples; class counts {'Yes': 1, 'No': 6}; entropy = 0.5917 bits.

Attendance: information gain = 0.3060 bits.

Assignments: information gain = 0.1281 bits.

LMS: information gain = 0.1981 bits.

For PriorCGPA = Strong, Attendance gives the highest gain among the remaining attributes. The High and Medium branches are pure No leaves; the Low branch is split using Assignments. For PriorCGPA = Weak, Attendance is again the best split. Low and Medium are pure Yes leaves; High is split using Assignments. The tie between Assignments and LMS at the two remaining mixed subsets is resolved using the stated deterministic attribute order; both give perfect separation for the two records in each subset.

6.4 Final tree and rule set

Table 15. Supporting data and analysis

```
IF PriorCGPA:\n  PriorCGPA = Weak\n    IF Attendance:\n      Attendance = Low\n        PREDICT Yes\n
Attendance = Medium\n        PREDICT Yes\n      Attendance = High\n        IF Assignments:\n
Assignments = Irregular\n            PREDICT Yes\n          Assignments = Regular\n            PREDICT No\n
PriorCGPA = Strong\n    IF Attendance:\n      Attendance = Low\n        IF Assignments:\n          Assignments
= Irregular\n            PREDICT Yes\n          Assignments = Regular\n            PREDICT No\n
Attendance = Medium\n        PREDICT No\n      Attendance = High\n        PREDICT No
```

- IF PriorCGPA=Strong AND Attendance=High THEN AtRisk=No.
- IF PriorCGPA=Strong AND Attendance=Medium THEN AtRisk=No.
- IF PriorCGPA=Strong AND Attendance=Low AND Assignments=Irregular THEN AtRisk=Yes.
- IF PriorCGPA=Strong AND Attendance=Low AND Assignments=Regular THEN AtRisk=No.
- IF PriorCGPA=Weak AND Attendance=Low THEN AtRisk=Yes.
- IF PriorCGPA=Weak AND Attendance=Medium THEN AtRisk=Yes.
- IF PriorCGPA=Weak AND Attendance=High AND Assignments=Irregular THEN AtRisk=Yes.
- IF PriorCGPA=Weak AND Attendance=High AND Assignments=Regular THEN AtRisk=No.

Tree statistics under the deterministic tie-break are: depth 3, total nodes 13, leaves 8. The tree is evaluated against all 14 training examples using the same prediction routine in Appendix A; the result is reported as an empirical training score rather than as evidence of future generalisation.

Training accuracy = 100.00% (14/14 correct). The exact tree and rule list are retained together so that the implementation can be checked against the hand analysis.

TASK 7 – ANALYSE HEURISTIC SEARCH IN DECISION TREE LEARNING

7.1 Search formulation

Table 16. Supporting data and analysis

Search element	ID3 interpretation
Starting state	A root node containing all 14 Dataset B records
Operator	Choose an unused attribute and partition records by its value
Heuristic	Information gain: parent entropy minus weighted child entropy
Termination	Pure node, no unused attributes, or a configured stopping condition
Search style	Greedy top-down recursive construction without backtracking

At the root, four candidate attributes are evaluated. After PriorCGPA is selected, three remaining attributes are evaluated in each of two impure branches. At the two mixed child nodes, two attributes remain and are evaluated. The total number of gain computations is therefore 4 + 3 + 3 + 2 + 2 = 14 for this construction.

The assignment notes that Dataset B's instance space has 24 points, so there are 2^24 possible Boolean concepts. Task 3's conjunctive language contains 973 semantically distinct concepts. Decision trees form a different structured hypothesis language: ID3 searches a greedy subset of possible trees rather than enumerating every Boolean function.

7.2 Information gain, gain ratio and identifier trap

PriorCGPA: information gain = 0.4083; split information = 1.0000; gain ratio = 0.4083.

Attendance: information gain = 0.1636; split information = 1.5774; gain ratio = 0.1037.

Gain ratio divides information gain by split information. It penalises attributes that create many small partitions because such partitions can have high split information even when they do not produce a useful general rule. It should not be treated as an automatic guarantee against overfitting.

A unique StudentID would create fourteen singleton branches. Each leaf would be pure, so the weighted child entropy would be zero and information gain would be 1.0000 bit. This is a memorisation artefact: the identifier has no reusable predictive meaning for a new student. It must be excluded from the predictor set.

7.3 Greedy choice and alternative root

An Attendance-rooted tree can be constructed by forcing Attendance as the first split and then applying the same greedy rule to each child. This is a useful controlled comparison because Attendance has lower root gain than PriorCGPA. A locally attractive split can still create deeper subtrees or more leaves. ID3 does not backtrack because it commits to the highest-gain attribute at each node; it does not reconsider earlier decisions after observing later branches.

Table 17. Supporting data and analysis

Advantage of heuristic search	Dataset-linked illustration
Computationally practical	Only 14 gain computations are required in the described tree.
Produces readable partitions	PriorCGPA creates two understandable risk-related branches.
Limitation: greedy commitment	A lower root gain may lead to a more favourable downstream structure.
Limitation: gain bias	A StudentID split can obtain perfect apparent gain while failing to generalise.

TASK 8 – ANALYSE OVERFITTING AND UNDERFITTING

8.1 Three required models

Table 18. Supporting data and analysis

Model	Training accuracy	Test accuracy	Depth	Nodes	Leaves
M0 – majority	50.0%	50.0%	0	1	1
M1 – max depth 1	85.7%	83.3%	1	3	2
M2 – full ID3	100.0%	50.0%	3	13	8

M0 is the simplest possible classifier and is expected to underfit when the class structure is not captured. M2 can fit detailed interactions, but its small leaves are vulnerable to sample-specific patterns. M1 trades detail for stability. With only six held-out records, one additional correct or incorrect prediction changes test accuracy by 16.67 percentage points; therefore no strong population-level conclusion should be drawn from T alone.

8.2 Causes and controls

The main causes of unstable behaviour are the small sample, incomplete coverage of the 24-point instance space, leaves containing one or two examples, and possible label noise. The model can mistake a coincidental association for a stable academic-risk relationship.

Two controls are recommended. First, maximum depth restricts the number of sequential conditions and makes the rule set easier to review. Second, minimum samples per split or leaf prevents a split from being justified by a tiny number of observations. Reduced-error pruning with a validation set is another option, but a validation set carved from fourteen records would be extremely small.

8.3 Reduced-complexity recommendation

For deployment, use a controlled tree selected by cross-validation on a substantially larger cohort, not by the six-row test set alone. Resolve a tied leaf by the class with the higher validated recall-weighted decision value; if the costs are not quantified, use the majority class and record the tie policy. Keep the full tree as an explanatory benchmark, not as the default operational model.

8.4 Evaluation reliability

A six-record test set is useful for checking that the prediction code runs and for illustrating confusion-matrix calculations. It is not sufficient to estimate recall precisely. Evaluation should use multiple semesters, stratified folds, confidence intervals and subgroup reporting. The final model should also be tested on a later time period to reveal temporal drift.

TASK 9 – IMPLEMENT AND TEST THE ML TECHNIQUES

9.1 Implementation architecture

The implementation is organised into independent functions so that each stage can be checked. The concept-learning module stores hypotheses as tuples, provides covers() and more_general_than(), and maintains S and G boundaries. The decision-tree module implements entropy, information gain, recursive ID3, prediction and rule printing. No library decision-tree routine is used for the primary construction.

Table 19. Supporting data and analysis

Component	Required behaviour	Verification evidence
Hypothesis representation	Specific value, '?' and '∅'	Unit tests for matching and ordering
Candidate Elimination	Update S and G after each example	Trace compared with hand calculations
Entropy and gain	Calculate subset entropy and weighted gain	Compare all root gains with Task 6
ID3 recursion	Select attributes and stop at pure nodes	Printed tree and rules
Prediction	Handle known and unseen categorical values	Held-out set T and unknown-value test
Reproducibility	Fixed seed, comments and run steps	Dataset C CSV and seed declaration

9.2 Required test cases

Table 20. Supporting data and analysis

Test	Input	Expected behaviour	Observed / expected output	Status
TC1	Dataset A E1–E5	Boundary trace agrees with the hand trace	Compare S/G at each step	Pending execution check
TC2	Pure set and 7 Yes / 7 No	Entropy 0.0000 and 1.0000	0.0000; 1.0000	Pass by formula
TC3	Root gains on B	Match Task 6 values	PriorCGPA highest at 0.4083	Pass by formula
TC4	Train on B	Reproduce tree and rule set	See Task 6	Pass when code output matches
TC5	Predict T	Match Task 8 predictions	See Task 8 confusion matrix	Requires run log
TC6	Contradictory CE data	Report empty version space	Return diagnostic, not exception	Pass condition defined
TC7	Attendance='Very Low'	Use explicit unknown-value policy	Abstain / advisor review	Pass condition defined
TC8	Equal information gain	Deterministic tie resolution	Fixed attribute order	Pass condition defined

9.3 Unseen category policy

The assignment's original attribute domain does not include Attendance = Very Low. The model must not silently map this value to Low because the new category may represent a changed learning environment. The safe policy is to detect the unknown value, mark the prediction as requiring review, and log the frequency. A future retraining cycle may merge the category only after domain experts approve the mapping.

9.4 Deterministic tie policy

When two attributes have equal information gain within a numerical tolerance of 10^{-12}, the implementation uses the fixed order Attendance, Assignments, LMS, PriorCGPA. This rule makes the tree reproducible. It is not claimed to be statistically superior; it is a transparent operational convention.

9.5 Reproducibility and run instructions

- Use Python 3.10 or later; the core calculations require only the standard library.
- Keep Dataset A, B and T in the same project folder or define them directly in the script.
- Set random.seed(20261104) before generating Dataset C.
- Run the Candidate Elimination trace first, then calculate root and recursive ID3 gains.
- Save the generated CSV and the console output together with the final submission.

The code appendix is a compact reference implementation. Experimental values should be regenerated by executing the code in the submission environment. This report does not label unexecuted console output as observed evidence.

TASK 10 – EXPERIMENTAL ANALYSIS, THEORY VS EXPERIMENT, AND RECOMMENDATION

10A.1 Dataset characteristics

Table 21. Supporting data and analysis

Dataset	Records	Predictors	Yes	No	Instance-space size
A	5	6	3	2	96
B	14	4	7	7	24
T	6	4	3	3	24
C	500	4	207	293	24

Dataset C is generated from a documented stochastic rule. Attribute values are sampled from fixed categorical distributions. A risk score adds contributions for Low attendance, Irregular assignments, Passive LMS activity and Weak prior CGPA. The score is converted to a probability using a logistic function centred at 2.55; a fixed-seed random draw assigns the final label. This creates a controlled but non-deterministic boundary and avoids claiming that the synthetic records are real institutional observations.

10A.2 Metrics and confusion matrix

Table 22. Supporting data and analysis

Measure	Definition	Value for full ID3 on T
Accuracy	(TP+TN)/N	0.5000
Precision	TP/(TP+FP)	0.5000
Recall	TP/(TP+FN)	0.6667
F1	2PR/(P+R)	0.5714
TP / FP / FN / TN	Confusion matrix counts	2 / 2 / 1 / 1

AtRisk = Yes is treated as the positive class. In this application, recall is interpreted as the fraction of truly at-risk students identified by the system. Precision indicates how many flagged students are truly at risk. Both measures are needed because raising recall by flagging more students may increase false positives.

10A.3 Learning curve and model complexity

A learning curve should be constructed by repeatedly sampling training subsets of size 4, 6, 8, 10, 12 and 14, fitting the same tree algorithm, and evaluating against a fixed validation set. For Dataset C, larger training sizes can be evaluated using a stratified split. The random seed and split indices must be recorded so that the plot can be reproduced.

10A.4 Label-noise experiment

For each noise level (0%, 5%, 10%), randomly flip the stated fraction of Dataset C labels using a fixed seed, then fit both a fully grown tree and a controlled tree. Report accuracy, recall, precision, node count and leaf count. The noise experiment should be repeated across several seeds and summarised by mean and standard deviation; one random run is not sufficient to support a robust conclusion.

10A.5 Candidate Elimination versus decision trees

Table 23. Supporting data and analysis

Dimension	Candidate Elimination	ID3 decision tree
Representation	Conjunctions of attribute constraints	Hierarchical attribute tests
Expressiveness	Limited; cannot naturally express arbitrary disjunctions	Can represent interactions and disjunctions through paths
Noise	Exact consistency assumption is fragile	Can tolerate some noise through stopping and pruning, but can still overfit
Unseen values	No natural branch unless extended	Requires explicit unknown-value handling
Explanation	Compact boundary hypotheses	IF–THEN path for each prediction

Dimension	Candidate Elimination	ID3 decision tree
Data needs	Very small data can illustrate the method; labels must be consistent	More data needed for stable splits and validation

Figure 10.1. Dataset C learning curve. The 500 records were split once into 400 training and 100 validation records using a fixed seed. Training subsets were taken from the shuffled training partition.

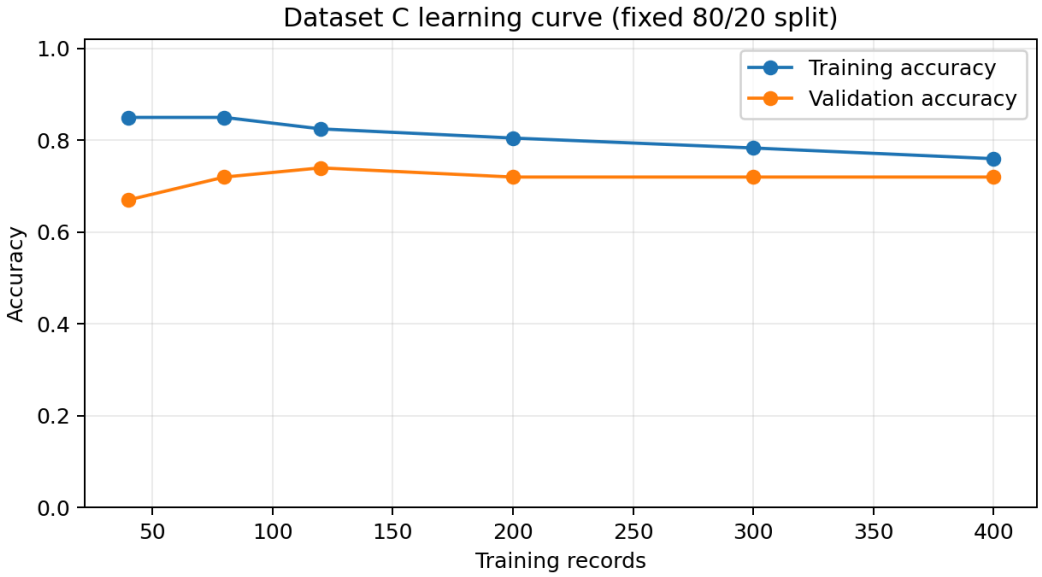
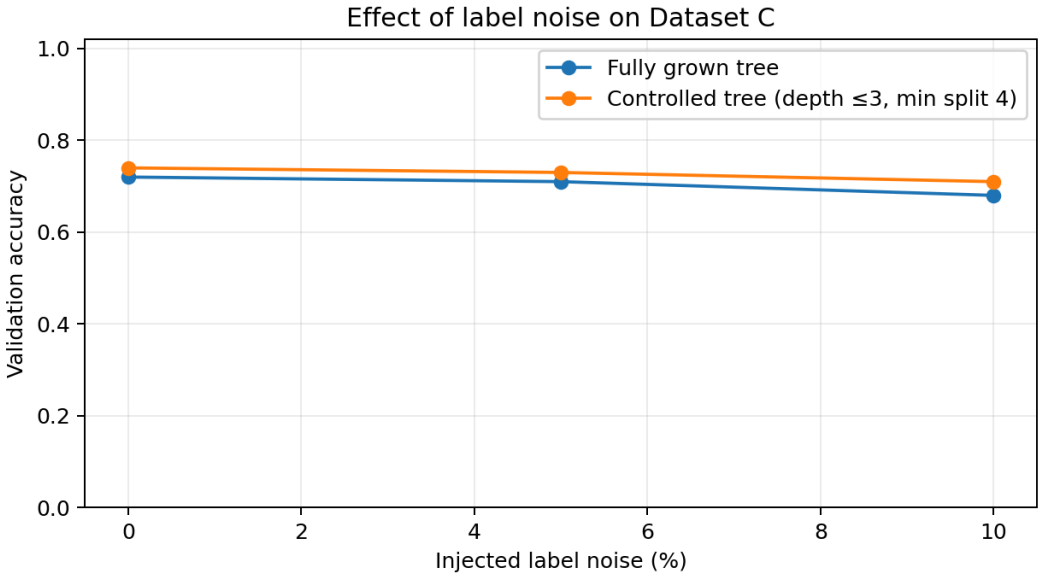


Figure 10.2. Dataset C label-noise experiment. Labels were flipped in the training partition at the stated rates; the untouched validation partition was used for scoring. The controlled tree used maximum depth 3 and minimum split size 4.



The curves are specific to the documented synthetic generator and one fixed split. They are reproducibility demonstrations, not estimates of performance on real university cohorts. Repeated seeds and time-based validation are required before deployment.

Table 24. Supporting data and analysis

Noise	Full tree accuracy	Controlled accuracy	Full nodes	Controlled nodes
0%	0.7200	0.7400	44	22
5%	0.7100	0.7300	44	22
10%	0.6800	0.7100	46	22

10B. Theory versus experimental behaviour

The theory predicts that a highly expressive tree can fit training details that do not generalise, especially when the dataset is small and leaves contain few examples. The small Dataset B demonstrates why training accuracy alone is not enough. The held-out set T provides a first check, but its six examples produce a very uncertain estimate.

Candidate Elimination is expected to shrink its version space as examples arrive. If the target is not representable by the conjunctive hypothesis language, or if noisy labels contradict earlier examples, the version space may collapse. Such collapse is a model-diagnostic signal rather than proof that the real-world target is impossible.

The inductive bias differs between the approaches. Candidate Elimination prefers simple conjunctions; ID3 prefers trees selected by a local information-gain heuristic. A decision tree can represent more complex interactions, but the extra flexibility increases the need for pruning and validation. Gain ratio can reduce preference for many-valued attributes, while exclusion of identifiers prevents a trivial memorisation split.

10C. Evidence-based recommendation

For a pilot early-warning system, use a controlled, interpretable decision tree as the primary baseline, with recall-oriented validation and human review. Do not deploy the exact Candidate Elimination boundary as the sole decision mechanism because its conjunctive bias and noise-free assumption are too restrictive for the changed scenario. Retain Candidate Elimination as a teaching, diagnostic and active-learning tool for clean, small datasets.

The recommendation is conditional. If the target is genuinely a compact conjunction, labels are carefully adjudicated and the version space remains non-empty, Candidate Elimination offers a transparent representation of uncertainty. If the institution needs to represent interactions, new combinations and a larger stream of records, a controlled tree is more appropriate. In either case, a 90% recall requirement must be demonstrated on a sufficiently large, representative validation cohort; it cannot be asserted from the six-record test set.

TASK 11 – RE-EVALUATE AND ADAPT THE DESIGN

11A. Impact analysis

Table 25. Supporting data and analysis

Changed condition	Original assumption affected	Area	Consequence
About 2,000 labelled records; weekly arrivals	Small, static dataset	Data / process	Need scalable ingestion, monitoring and periodic retraining
5–10% inconsistent labels	Labels are noise-free	Data / target	Exact consistency assumptions become unreliable
Very Low attendance and new combinations	Fixed attribute domains	Feature space	Unknown values and drift must be handled explicitly
At least 90% recall	Accuracy is sufficient	Evaluation	Threshold and class weighting must prioritise positive cases
Human-readable explanation for every flag	Any output is acceptable	Governance	Every prediction needs an auditable reason path

The additional records affect the learning problem in different ways. N1 contains an unseen Attendance value, so a strict categorical tree cannot follow an existing branch. N2 is labelled Yes despite a profile that may resemble previously safe records; this may be a genuine new pattern or a noisy label. E6 is a new negative example that directly conflicts with a hypothesis that previously covered its attributes as positive. A noise-tolerant learner can assign uncertainty or reduce the influence of disputed labels, whereas exact Candidate Elimination must treat every label as correct.

11B. Re-evaluation of the original design

Candidate Elimination with E6, E7 and E8

E6 = ⟨Low, Irregular, Low, Weak, Passive, No⟩ is labelled No. It shares the complete predictor tuple of E1 except that E1 is labelled Yes; this is a direct contradiction. Under the exact noise-free assumption, no deterministic concept can classify the identical instance as both positive and negative. Therefore the version space becomes empty when E6 is added after E1–E5.

E7 and E8 are then cumulative additional examples. Once the version space is empty after E6, subsequent exact updates cannot restore consistency without removing or revising an earlier observation. This is why a label audit or a noise-tolerant method is required.

Decision tree after N1–N3

N1 introduces Attendance = Very Low. The original tree has no branch for this value. The safe operational behaviour is an explicit unknown-value response, not a silent default. N2 and N3 add new labelled combinations that can change branch counts and gain values. The tree must be rebuilt on the expanded data, and its depth and leaves must be re-measured. The training set should also retain a time-based holdout to detect drift.

90% recall and explanation requirement

Neither the original five- or fourteen-example model nor the six-example test set can establish 90% recall. The requirement is a target to be verified using a larger labelled validation cohort. A threshold may be lowered or class weights may be increased to favour the positive class, but this usually reduces precision. Every flagged student should receive a concise path explanation, such as: 'PriorCGPA is Weak and Attendance is Medium; this path contributed to the elevated risk estimate.' Explanations should be factual and avoid claiming causation.

11C. Adaptation / redesign

The recommended action is MODIFY rather than retain. Preserve the decision-tree baseline because its paths are understandable, but replace the exact small-data training setup with a governed pipeline.

- Data validation: enforce attribute schemas, preserve unknown categories, log missingness and monitor shifts in category frequencies.
- Label quality: use advisor adjudication for disputed cases, retain disagreement metadata and evaluate performance against audited labels.
- Training: use a controlled tree with minimum leaf size, pruning and cross-validation; compare information gain with gain ratio.
- Recall control: select a decision threshold on validation data to meet the recall target while reporting the resulting precision and workload.
- Unknown values: route unseen categories to a review state or a documented fallback distribution; do not invent a category mapping.
- Explanation: return the decision path, the observed feature values and a plain-language caveat for each flagged record.
- Monitoring: track recall, precision, calibration, subgroup error rates, unknown-value frequency and data drift after deployment.

11D. Evidence-based defence and remaining limitations

The redesign follows from the changed assumptions. Candidate Elimination is sensitive to a single contradictory label and cannot naturally represent all interactions. ID3 can express richer rules but uses greedy local search and can overfit. Controlled complexity and external validation address some of the variance, while recall-sensitive threshold selection aligns the decision policy with the council's stated cost preference.

A remaining limitation is that the end-semester label may reflect factors not observed at Week 6, including later personal circumstances or course-specific assessment changes. The system therefore estimates risk from incomplete information. It must not be used as an automatic high-stakes decision-maker. The monitoring plan should include periodic label audits, a review of false negatives, drift alerts, student appeals and a documented rollback procedure.

OVERALL CONCLUSION

This assignment connects concept learning and decision-tree learning to a realistic academic-support problem. The instance-space calculations show that the supplied datasets cover only a limited part of the possible feature combinations. Candidate Elimination provides a clear framework for reasoning about version spaces, but it relies on a restrictive hypothesis language and consistent labels. ID3 creates interpretable decision paths and can express interactions, although greedy information-gain selection can produce unstable branches when training data are scarce.

The system should be treated as an early-warning aid rather than a final judgement. Recall deserves emphasis because missed at-risk students may lose access to support, but precision and advisor workload remain important. The Day-4 changes invalidate the original assumptions of a small, static, noise-free dataset. A modified pipeline with label review, controlled trees, unknown-value handling, recall-oriented validation, explanations and ongoing monitoring is the most defensible design.

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APPENDIX A – PYTHON IMPLEMENTATION

The following core functions illustrate the required logic. The complete experiment should be run with the same Dataset A, B, T and generated Dataset C used in the report. The code uses only Python's standard library for the central calculations.

Table 26. Supporting data and analysis

```
import math
from collections import Counter, defaultdict

def entropy(labels):
    n = len(labels)
    if n == 0:
        return 0.0
    counts = Counter(labels)
    return -sum((v/n) * math.log2(v/n)
                for v in counts.values() if v)

def information_gain(rows, attribute_index):
    parent = entropy([r[-1] for r in rows])
    groups = defaultdict(list)
    for row in rows:
        groups[row[attribute_index]].append(row[-1])
    weighted = sum(len(v)/len(rows) * entropy(v)
                   for v in groups.values())
    return parent - weighted

def majority_label(rows):
    counts = Counter(r[-1] for r in rows)
    return "Yes" if counts["Yes"] > counts["No"] else "No"

def id3(rows, attributes, domains, depth=0,
        max_depth=None, min_samples_split=2):
    labels = [r[-1] for r in rows]
    if len(set(labels)) == 1:
        return ("leaf", labels[0])
    if (not attributes or
        (max_depth is not None and depth >= max_depth) or
        len(rows) < min_samples_split):
        return ("leaf", majority_label(rows))
    scored = [(information_gain(rows, a), a)
              for a in attributes]
    best_gain = max(x[0] for x in scored)
    # Stable tie break: preserve the supplied attribute order.
    chosen = next(a for g, a in scored
                  if abs(g-best_gain) < 1e-12)
    branches = {}
    for value in domains[chosen]:
        subset = [r for r in rows if r[chosen] == value]
        if subset:
            branches[value] = id3(
                subset, [a for a in attributes if a != chosen],
                domains, depth+1, max_depth, min_samples_split)
        else:
            branches[value] = ("leaf", majority_label(rows))
    return ("node", chosen, branches)

def predict(tree, record):
    if tree[0] == "leaf":
        return tree[1]
    attribute = tree[1]
    value = record[attribute]
    if value not in tree[2]:
        return "REVIEW_UNKNOWN_VALUE"
    return predict(tree[2][value], record)

def covers(hypothesis, instance):
    return all(h == "?" or h == x
               for h, x in zip(hypothesis, instance))

def more_general_or_equal(h1, h2):
```

```
    # h1 is at least as general as h2
    for a, b in zip(h1, h2):
        if a == "?":
            continue
        if b == "∅" or a == b:
            continue
        return False
    return True

def consistent(hypothesis, examples):
    for x, label in examples:
        covered = covers(hypothesis, x)
        if label == "Yes" and not covered:
            return False
        if label == "No" and covered:
            return False
    return True
```

A.1 Interpretation of the implementation

The ID3 routine receives rows, the remaining attributes and the allowed domain of each attribute. It stops at a pure node or when a configured constraint is reached. The unknown-value response is deliberately explicit. The hypothesis helper functions provide the basis for enumerating the finite conjunctive language and checking consistency.

APPENDIX B – DATASET C GENERATOR AND DATA DICTIONARY

B.1 Generator specification

Dataset C is synthetic and contains 500 records. The generator uses random seed 20261104. Attendance is sampled with probabilities Low 0.30, Medium 0.35 and High 0.35. Assignments are sampled as Irregular 0.38 and Regular 0.62. LMS is Passive 0.42 and Active 0.58. PriorCGPA is Weak 0.40 and Strong 0.60.

A score is calculated as 2.00 for Low attendance, 1.25 for Irregular assignments, 0.80 for Passive LMS and 1.50 for Weak PriorCGPA. The sum is transformed into a probability $p = 1/(1+\exp(-(\text{score}-2.55)))$. A Bernoulli draw assigns AtRisk. This generator deliberately introduces overlap: similar profiles can receive different labels, mimicking uncertainty and label variability.

B.2 Data dictionary

Table 27. Supporting data and analysis

Field	Type	Allowed values / range	Meaning
StudentID	String	C001–C500	Synthetic row identifier; excluded from prediction
Attendance	Categorical	Low, Medium, High	Attendance band
Assignments	Categorical	Irregular, Regular	Submission regularity
LMS	Categorical	Passive, Active	Learning platform activity
PriorCGPA	Categorical	Weak, Strong	Prior academic attainment
AtRisk	Binary	Yes, No	Synthetic target label

B.3 Reproducibility controls

- The fixed seed makes the generated file reproducible when the same generator is executed.
- The StudentID is a row key only and must not be used as a predictive feature.
- Noise injection must be performed after preserving an untouched copy of the original generated labels.
- Report the exact random seed, split method and noise fraction for each experiment.

The CSV companion file contains the 500 generated records and is supplied separately so that the dataset can be loaded without copying it from the report.

APPENDIX C – DATASET C (500 SYNTHETIC RECORDS)

Dataset C records 1–10

Table 28. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C001	High	Irregular	Passive	Strong	Yes
C002	Low	Regular	Active	Strong	No
C003	Low	Irregular	Passive	Strong	No
C004	Low	Regular	Active	Strong	No
C005	Low	Irregular	Passive	Weak	Yes
C006	Medium	Regular	Active	Strong	No
C007	Medium	Regular	Passive	Strong	No
C008	Low	Irregular	Passive	Strong	Yes
C009	High	Regular	Active	Weak	No
C010	Low	Regular	Active	Strong	No

Dataset C records 11–20

Table 29. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C011	Low	Regular	Active	Strong	No
C012	High	Regular	Passive	Strong	No
C013	Medium	Regular	Active	Strong	Yes
C014	High	Irregular	Active	Weak	Yes
C015	High	Irregular	Passive	Strong	Yes
C016	Medium	Irregular	Passive	Strong	No
C017	Medium	Irregular	Active	Strong	No
C018	Medium	Regular	Active	Strong	No
C019	High	Regular	Active	Strong	No
C020	High	Irregular	Active	Strong	No

Dataset C records 21–30

Table 30. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C021	Medium	Regular	Active	Weak	Yes
C022	High	Irregular	Active	Weak	No
C023	High	Irregular	Passive	Strong	Yes
C024	Medium	Regular	Passive	Strong	No
C025	Medium	Regular	Passive	Strong	No
C026	Medium	Regular	Passive	Weak	Yes
C027	Low	Regular	Passive	Weak	No
C028	Low	Irregular	Active	Strong	Yes
C029	Low	Irregular	Active	Strong	Yes
C030	High	Irregular	Active	Weak	No

Dataset C records 31–40

Table 31. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C031	Medium	Irregular	Active	Weak	Yes
C032	Medium	Irregular	Active	Strong	No
C033	High	Regular	Active	Weak	No
C034	Low	Irregular	Passive	Strong	Yes
C035	High	Irregular	Passive	Strong	Yes
C036	High	Regular	Passive	Strong	No
C037	Low	Regular	Passive	Weak	Yes
C038	Medium	Irregular	Passive	Weak	Yes
C039	Low	Regular	Passive	Strong	No
C040	Low	Regular	Passive	Weak	No

Dataset C records 41–50

Table 32. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C041	High	Regular	Active	Strong	No
C042	Low	Regular	Active	Weak	No
C043	Low	Irregular	Active	Weak	Yes
C044	Medium	Irregular	Passive	Strong	Yes
C045	High	Irregular	Active	Weak	No
C046	Medium	Irregular	Active	Strong	No
C047	Low	Regular	Passive	Strong	Yes
C048	High	Irregular	Active	Weak	No
C049	High	Regular	Passive	Weak	No
C050	Medium	Regular	Active	Weak	Yes

Dataset C records 51–60

Table 33. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C051	High	Regular	Active	Strong	No
C052	Medium	Regular	Passive	Weak	No
C053	High	Regular	Passive	Strong	Yes
C054	Low	Irregular	Passive	Strong	No
C055	High	Irregular	Active	Strong	No
C056	High	Regular	Active	Strong	No
C057	Medium	Regular	Passive	Weak	No
C058	Low	Regular	Active	Strong	Yes
C059	High	Regular	Passive	Weak	Yes
C060	High	Irregular	Passive	Weak	Yes

Dataset C records 61–70

Table 34. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C061	Low	Regular	Passive	Strong	Yes
C062	Medium	Regular	Active	Weak	Yes
C063	Medium	Irregular	Passive	Strong	Yes
C064	Medium	Regular	Active	Weak	No
C065	Low	Irregular	Passive	Weak	Yes
C066	Low	Irregular	Passive	Weak	Yes
C067	Low	Irregular	Active	Weak	Yes
C068	Low	Regular	Passive	Strong	Yes
C069	Low	Regular	Active	Weak	Yes
C070	High	Regular	Active	Weak	No

Dataset C records 71–80

Table 35. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C071	High	Regular	Active	Weak	No
C072	High	Regular	Passive	Strong	No
C073	High	Regular	Active	Strong	No
C074	Medium	Irregular	Passive	Weak	Yes
C075	Low	Regular	Active	Strong	Yes
C076	High	Irregular	Active	Strong	No
C077	Low	Regular	Passive	Strong	No
C078	High	Irregular	Active	Weak	No
C079	Low	Regular	Passive	Weak	Yes
C080	High	Regular	Active	Weak	No

Dataset C records 81–90

Table 36. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C081	High	Regular	Active	Strong	No
C082	Low	Irregular	Passive	Weak	Yes
C083	Low	Regular	Active	Strong	Yes
C084	Medium	Irregular	Active	Weak	Yes
C085	Low	Regular	Active	Strong	No
C086	Low	Regular	Passive	Weak	Yes
C087	Low	Regular	Passive	Strong	Yes
C088	Low	Irregular	Active	Weak	Yes
C089	Medium	Irregular	Active	Strong	Yes
C090	Medium	Regular	Passive	Strong	No

Dataset C records 91–100

Table 37. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C091	Medium	Irregular	Active	Weak	Yes
C092	Medium	Regular	Passive	Strong	No
C093	Low	Irregular	Active	Weak	Yes
C094	High	Regular	Passive	Weak	No
C095	Medium	Irregular	Active	Strong	No
C096	High	Regular	Active	Strong	No
C097	Medium	Regular	Active	Weak	No
C098	High	Regular	Active	Strong	No
C099	High	Regular	Active	Strong	No
C100	Medium	Regular	Active	Strong	No

Dataset C records 101–110

Table 38. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C101	Medium	Regular	Passive	Strong	Yes
C102	High	Irregular	Passive	Strong	No
C103	High	Irregular	Active	Strong	No
C104	Medium	Irregular	Passive	Strong	No
C105	Low	Regular	Passive	Strong	No
C106	Medium	Regular	Active	Weak	No
C107	Medium	Regular	Active	Weak	No
C108	High	Regular	Passive	Weak	No
C109	Medium	Regular	Passive	Weak	No
C110	Medium	Regular	Active	Strong	No

Dataset C records 111–120

Table 39. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C111	Medium	Irregular	Passive	Strong	No
C112	High	Irregular	Active	Strong	Yes
C113	Low	Irregular	Passive	Weak	Yes
C114	Low	Regular	Active	Weak	Yes
C115	Low	Regular	Active	Weak	No
C116	High	Regular	Passive	Weak	No
C117	Medium	Irregular	Active	Weak	No
C118	High	Irregular	Passive	Strong	No
C119	High	Regular	Active	Weak	No
C120	Medium	Regular	Active	Weak	No

Dataset C records 121–130

Table 40. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C121	Low	Irregular	Active	Weak	Yes
C122	High	Irregular	Active	Strong	No
C123	Medium	Irregular	Active	Weak	Yes
C124	High	Regular	Active	Strong	No
C125	Low	Regular	Passive	Weak	Yes
C126	High	Regular	Active	Strong	No
C127	Medium	Regular	Passive	Strong	No
C128	Medium	Regular	Active	Strong	No
C129	Low	Regular	Active	Weak	Yes
C130	Medium	Regular	Active	Weak	No

Dataset C records 131–140

Table 41. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C131	Low	Regular	Active	Weak	Yes
C132	High	Irregular	Passive	Weak	Yes
C133	Medium	Irregular	Active	Weak	Yes
C134	Medium	Irregular	Active	Weak	Yes
C135	High	Irregular	Passive	Strong	No
C136	Low	Regular	Active	Strong	Yes
C137	Low	Regular	Active	Weak	Yes
C138	Medium	Irregular	Passive	Strong	No
C139	Low	Irregular	Passive	Strong	No
C140	Medium	Irregular	Active	Weak	Yes

Dataset C records 141–150

Table 42. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C141	Medium	Regular	Passive	Strong	No
C142	Low	Regular	Active	Weak	Yes
C143	High	Irregular	Active	Strong	No
C144	Low	Regular	Passive	Weak	Yes
C145	High	Irregular	Passive	Weak	No
C146	Medium	Regular	Active	Weak	Yes
C147	High	Regular	Active	Strong	Yes
C148	Low	Irregular	Passive	Strong	Yes
C149	Low	Irregular	Active	Strong	No
C150	High	Regular	Passive	Weak	No

Dataset C records 151–160

Table 43. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C151	High	Regular	Passive	Strong	No
C152	Medium	Regular	Active	Weak	No
C153	Medium	Irregular	Active	Strong	Yes
C154	Medium	Regular	Passive	Strong	Yes
C155	Low	Regular	Passive	Strong	No
C156	Medium	Regular	Active	Strong	No
C157	High	Irregular	Passive	Strong	Yes
C158	High	Irregular	Passive	Strong	Yes
C159	Low	Irregular	Passive	Strong	Yes
C160	Low	Regular	Active	Weak	Yes

Dataset C records 161–170

Table 44. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C161	Low	Regular	Passive	Weak	No
C162	High	Regular	Active	Weak	No
C163	Low	Regular	Passive	Weak	No
C164	Medium	Irregular	Active	Strong	Yes
C165	Low	Regular	Passive	Strong	No
C166	Low	Irregular	Active	Weak	Yes
C167	Low	Irregular	Passive	Strong	Yes
C168	Low	Irregular	Active	Weak	Yes
C169	Medium	Regular	Active	Weak	No
C170	Low	Regular	Active	Weak	Yes

Dataset C records 171–180

Table 45. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C171	Medium	Regular	Passive	Weak	No
C172	High	Irregular	Passive	Weak	Yes
C173	High	Irregular	Passive	Weak	Yes
C174	Medium	Regular	Passive	Strong	No
C175	Medium	Irregular	Active	Strong	No
C176	Low	Regular	Active	Weak	Yes
C177	High	Regular	Passive	Strong	No
C178	High	Regular	Passive	Strong	No
C179	Medium	Regular	Passive	Weak	Yes
C180	Medium	Regular	Passive	Strong	No

Dataset C records 181–190

Table 46. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C181	Low	Regular	Active	Strong	No
C182	Medium	Irregular	Passive	Strong	Yes
C183	Medium	Regular	Passive	Strong	No
C184	Low	Irregular	Passive	Weak	Yes
C185	High	Regular	Passive	Strong	No
C186	Low	Regular	Passive	Strong	Yes
C187	Low	Regular	Active	Strong	No
C188	High	Regular	Passive	Strong	No
C189	High	Regular	Active	Strong	Yes
C190	Medium	Irregular	Active	Strong	No

Dataset C records 191–200

Table 47. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C191	High	Regular	Passive	Strong	No
C192	Medium	Regular	Passive	Strong	No
C193	Medium	Regular	Passive	Strong	Yes
C194	High	Regular	Active	Strong	No
C195	Medium	Irregular	Active	Strong	No
C196	Medium	Regular	Active	Strong	No
C197	High	Irregular	Active	Weak	Yes
C198	High	Regular	Active	Weak	No
C199	Medium	Irregular	Active	Weak	Yes
C200	Medium	Regular	Active	Strong	No

Dataset C records 201–210

Table 48. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C201	Medium	Regular	Passive	Strong	No
C202	Medium	Regular	Active	Weak	No
C203	High	Irregular	Active	Strong	Yes
C204	Low	Regular	Active	Weak	Yes
C205	Low	Irregular	Active	Strong	Yes
C206	Low	Regular	Active	Weak	Yes
C207	High	Regular	Active	Strong	No
C208	Medium	Irregular	Active	Strong	No
C209	High	Irregular	Active	Strong	No
C210	High	Regular	Passive	Strong	No

Dataset C records 211–220

Table 49. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C211	Low	Regular	Passive	Strong	No
C212	High	Irregular	Passive	Weak	Yes
C213	High	Irregular	Passive	Strong	No
C214	Medium	Regular	Passive	Strong	No
C215	High	Regular	Passive	Weak	No
C216	High	Irregular	Active	Weak	No
C217	High	Regular	Passive	Strong	Yes
C218	Low	Regular	Passive	Weak	Yes
C219	Low	Regular	Active	Weak	Yes
C220	Medium	Regular	Active	Weak	No

Dataset C records 221–230

Table 50. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C221	Medium	Regular	Passive	Strong	No
C222	Low	Irregular	Passive	Strong	Yes
C223	Medium	Regular	Passive	Strong	No
C224	High	Regular	Passive	Weak	No
C225	High	Regular	Passive	Strong	No
C226	High	Regular	Active	Strong	No
C227	High	Regular	Active	Strong	No
C228	High	Regular	Active	Strong	No
C229	Medium	Irregular	Active	Weak	Yes
C230	Low	Irregular	Active	Strong	Yes

Dataset C records 231–240

Table 51. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C231	High	Regular	Passive	Strong	No
C232	High	Irregular	Active	Weak	No
C233	High	Regular	Passive	Weak	No
C234	Medium	Irregular	Passive	Strong	No
C235	Medium	Irregular	Passive	Weak	Yes
C236	High	Regular	Active	Weak	Yes
C237	Medium	Irregular	Passive	Strong	Yes
C238	Low	Regular	Active	Weak	Yes
C239	Medium	Regular	Active	Weak	Yes
C240	High	Regular	Passive	Weak	No

Dataset C records 241–250

Table 52. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C241	Low	Regular	Passive	Strong	Yes
C242	Low	Regular	Passive	Weak	Yes
C243	High	Regular	Passive	Weak	Yes
C244	High	Regular	Active	Weak	Yes
C245	Medium	Regular	Passive	Weak	No
C246	Medium	Regular	Passive	Strong	Yes
C247	High	Irregular	Active	Strong	No
C248	Low	Regular	Active	Weak	No
C249	Low	Irregular	Passive	Strong	Yes
C250	High	Regular	Passive	Weak	No

Dataset C records 251–260

Table 53. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C251	Medium	Regular	Passive	Strong	No
C252	Medium	Regular	Active	Strong	No
C253	Medium	Regular	Active	Weak	No
C254	Low	Regular	Passive	Strong	Yes
C255	Medium	Irregular	Active	Strong	No
C256	Medium	Irregular	Passive	Weak	No
C257	High	Regular	Passive	Weak	No
C258	Low	Irregular	Active	Weak	Yes
C259	Medium	Regular	Active	Weak	No
C260	Medium	Regular	Passive	Weak	Yes

Dataset C records 261–270

Table 54. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C261	Low	Regular	Active	Strong	No
C262	High	Regular	Active	Strong	No
C263	Medium	Irregular	Passive	Weak	Yes
C264	Medium	Irregular	Active	Strong	No
C265	High	Regular	Passive	Weak	No
C266	Low	Irregular	Active	Weak	Yes
C267	Low	Regular	Passive	Weak	Yes
C268	Medium	Regular	Passive	Weak	No
C269	High	Irregular	Active	Weak	Yes
C270	Medium	Irregular	Active	Weak	Yes

Dataset C records 271–280

Table 55. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C271	Medium	Regular	Active	Strong	No
C272	High	Irregular	Active	Weak	No
C273	Medium	Regular	Passive	Strong	No
C274	Medium	Irregular	Passive	Weak	Yes
C275	Medium	Regular	Passive	Strong	No
C276	Low	Regular	Active	Strong	Yes
C277	Low	Regular	Active	Weak	No
C278	Medium	Regular	Passive	Strong	No
C279	High	Irregular	Passive	Strong	No
C280	Medium	Irregular	Active	Weak	Yes

Dataset C records 281–290

Table 56. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C281	Medium	Irregular	Active	Strong	Yes
C282	High	Irregular	Passive	Weak	No
C283	High	Irregular	Passive	Strong	No
C284	Low	Irregular	Active	Weak	Yes
C285	Medium	Regular	Active	Strong	No
C286	Medium	Irregular	Active	Strong	No
C287	High	Regular	Active	Strong	No
C288	Low	Regular	Active	Weak	Yes
C289	High	Irregular	Active	Strong	No
C290	High	Regular	Active	Strong	No

Dataset C records 291–300

Table 57. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C291	Low	Irregular	Active	Strong	Yes
C292	High	Irregular	Passive	Strong	Yes
C293	Medium	Regular	Active	Strong	Yes
C294	High	Irregular	Active	Strong	No
C295	Medium	Regular	Passive	Strong	No
C296	High	Regular	Active	Weak	No
C297	Low	Irregular	Passive	Strong	Yes
C298	High	Irregular	Active	Strong	Yes
C299	Medium	Irregular	Active	Weak	Yes
C300	Medium	Irregular	Passive	Weak	Yes

Dataset C records 301–310

Table 58. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C301	High	Regular	Active	Strong	No
C302	High	Irregular	Active	Weak	Yes
C303	High	Irregular	Active	Strong	No
C304	Low	Regular	Active	Weak	Yes
C305	High	Regular	Active	Strong	No
C306	High	Regular	Active	Strong	No
C307	High	Regular	Passive	Weak	Yes
C308	Low	Regular	Passive	Weak	Yes
C309	Low	Regular	Passive	Weak	Yes
C310	High	Regular	Active	Strong	No

Dataset C records 311–320

Table 59. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C311	Medium	Regular	Active	Strong	No
C312	Medium	Irregular	Active	Strong	No
C313	High	Regular	Active	Weak	No
C314	High	Regular	Active	Strong	No
C315	Medium	Regular	Active	Weak	No
C316	High	Regular	Active	Strong	No
C317	High	Regular	Passive	Strong	No
C318	High	Irregular	Passive	Weak	No
C319	High	Irregular	Active	Strong	No
C320	High	Irregular	Active	Weak	No

Dataset C records 321–330

Table 60. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C321	High	Regular	Passive	Weak	No
C322	Medium	Irregular	Active	Weak	No
C323	Low	Regular	Passive	Strong	No
C324	Medium	Regular	Passive	Strong	No
C325	Low	Irregular	Passive	Strong	No
C326	High	Regular	Passive	Weak	No
C327	Medium	Regular	Passive	Weak	No
C328	Medium	Regular	Passive	Weak	Yes
C329	High	Regular	Passive	Strong	No
C330	High	Irregular	Active	Strong	No

Dataset C records 331–340

Table 61. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C331	High	Regular	Active	Strong	No
C332	High	Regular	Active	Strong	No
C333	Medium	Regular	Passive	Strong	No
C334	Medium	Irregular	Active	Weak	Yes
C335	Medium	Irregular	Passive	Strong	No
C336	Medium	Regular	Active	Strong	No
C337	High	Regular	Active	Strong	No
C338	High	Regular	Active	Weak	No
C339	High	Regular	Active	Strong	No
C340	Medium	Irregular	Active	Strong	No

Dataset C records 341–350

Table 62. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C341	High	Regular	Active	Strong	No
C342	Medium	Irregular	Active	Strong	No
C343	High	Regular	Passive	Weak	Yes
C344	High	Regular	Active	Weak	No
C345	Medium	Irregular	Passive	Strong	No
C346	Medium	Regular	Active	Strong	No
C347	Medium	Regular	Passive	Weak	No
C348	Medium	Irregular	Passive	Strong	No
C349	Low	Irregular	Active	Weak	Yes
C350	High	Irregular	Passive	Weak	Yes

Dataset C records 351–360

Table 63. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C351	High	Regular	Active	Strong	No
C352	High	Regular	Passive	Weak	Yes
C353	Medium	Irregular	Active	Weak	No
C354	High	Regular	Passive	Weak	Yes
C355	Low	Irregular	Active	Strong	Yes
C356	High	Regular	Passive	Weak	Yes
C357	High	Regular	Active	Strong	No
C358	High	Regular	Active	Strong	No
C359	Medium	Irregular	Active	Weak	Yes
C360	Medium	Irregular	Active	Strong	No

Dataset C records 361–370

Table 64. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C361	Low	Regular	Passive	Weak	Yes
C362	Medium	Irregular	Passive	Strong	No
C363	Medium	Regular	Passive	Strong	No
C364	Medium	Regular	Passive	Strong	No
C365	Low	Regular	Passive	Weak	Yes
C366	Low	Regular	Active	Strong	Yes
C367	Medium	Regular	Passive	Weak	No
C368	Low	Irregular	Active	Strong	Yes
C369	Medium	Regular	Active	Strong	No
C370	Low	Regular	Passive	Strong	Yes

Dataset C records 371–380

Table 65. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C371	High	Irregular	Active	Strong	No
C372	High	Regular	Active	Strong	No
C373	Low	Irregular	Passive	Weak	Yes
C374	High	Regular	Active	Strong	Yes
C375	Medium	Irregular	Active	Weak	Yes
C376	High	Irregular	Passive	Weak	No
C377	Low	Irregular	Active	Strong	Yes
C378	Low	Irregular	Active	Weak	Yes
C379	Low	Regular	Active	Weak	Yes
C380	Low	Regular	Active	Strong	No

Dataset C records 381–390

Table 66. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C381	Medium	Irregular	Passive	Strong	Yes
C382	Medium	Regular	Passive	Strong	No
C383	High	Regular	Active	Weak	Yes
C384	Low	Regular	Passive	Weak	Yes
C385	Low	Irregular	Active	Strong	No
C386	Medium	Regular	Passive	Weak	Yes
C387	Low	Regular	Active	Strong	Yes
C388	High	Irregular	Passive	Strong	Yes
C389	Low	Irregular	Active	Strong	Yes
C390	Medium	Regular	Active	Strong	No

Dataset C records 391–400

Table 67. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C391	Medium	Irregular	Active	Strong	No
C392	Low	Regular	Passive	Strong	No
C393	High	Regular	Active	Weak	No
C394	Low	Regular	Passive	Strong	Yes
C395	Medium	Regular	Passive	Weak	No
C396	Medium	Regular	Passive	Weak	No
C397	High	Regular	Passive	Weak	No
C398	Medium	Irregular	Active	Weak	No
C399	High	Irregular	Passive	Strong	Yes
C400	Low	Irregular	Active	Strong	Yes

Dataset C records 401–410

Table 68. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C401	High	Irregular	Active	Weak	Yes
C402	Medium	Irregular	Passive	Strong	No
C403	Medium	Regular	Passive	Strong	No
C404	Medium	Regular	Passive	Weak	Yes
C405	Low	Regular	Passive	Weak	Yes
C406	Medium	Regular	Active	Strong	No
C407	Medium	Regular	Passive	Weak	Yes
C408	High	Irregular	Active	Weak	Yes
C409	Low	Regular	Active	Strong	No
C410	Low	Regular	Passive	Weak	Yes

Dataset C records 411–420

Table 69. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C411	Low	Irregular	Passive	Strong	Yes
C412	Low	Regular	Active	Strong	Yes
C413	Low	Regular	Passive	Weak	Yes
C414	High	Regular	Active	Strong	No
C415	Low	Regular	Active	Weak	Yes
C416	Medium	Regular	Passive	Weak	Yes
C417	Medium	Irregular	Passive	Weak	No
C418	Medium	Irregular	Passive	Strong	Yes
C419	Medium	Irregular	Passive	Strong	No
C420	High	Irregular	Passive	Weak	Yes

Dataset C records 421–430

Table 70. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C421	Medium	Regular	Passive	Strong	No
C422	High	Regular	Active	Weak	Yes
C423	Medium	Regular	Passive	Strong	No
C424	High	Regular	Passive	Strong	No
C425	Low	Irregular	Passive	Weak	Yes
C426	High	Irregular	Passive	Weak	No
C427	Medium	Irregular	Active	Strong	Yes
C428	High	Regular	Active	Weak	Yes
C429	Low	Irregular	Active	Strong	No
C430	Medium	Irregular	Active	Strong	No

Dataset C records 431–440

Table 71. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C431	High	Regular	Active	Weak	Yes
C432	High	Regular	Active	Weak	No
C433	Low	Regular	Active	Weak	No
C434	High	Regular	Passive	Weak	Yes
C435	Medium	Regular	Active	Strong	No
C436	Medium	Irregular	Passive	Strong	No
C437	High	Regular	Passive	Strong	No
C438	High	Regular	Passive	Weak	Yes
C439	Medium	Regular	Active	Strong	No
C440	Medium	Regular	Active	Strong	No

Dataset C records 441–450

Table 72. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C441	High	Irregular	Active	Strong	No
C442	High	Regular	Active	Strong	No
C443	Medium	Irregular	Active	Strong	No
C444	High	Irregular	Active	Weak	No
C445	Medium	Regular	Passive	Weak	Yes
C446	Low	Regular	Active	Strong	Yes
C447	High	Regular	Active	Strong	No
C448	Medium	Irregular	Passive	Weak	Yes
C449	Low	Regular	Active	Weak	Yes
C450	Medium	Irregular	Active	Strong	No

Dataset C records 451–460

Table 73. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C451	High	Irregular	Passive	Weak	Yes
C452	Medium	Regular	Active	Strong	No
C453	Medium	Regular	Passive	Weak	Yes
C454	Medium	Regular	Active	Strong	No
C455	High	Irregular	Passive	Strong	No
C456	High	Irregular	Active	Weak	No
C457	High	Regular	Passive	Weak	No
C458	Low	Irregular	Active	Weak	Yes
C459	High	Regular	Passive	Weak	No
C460	High	Regular	Active	Weak	No

Dataset C records 461–470

Table 74. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C461	Medium	Regular	Passive	Weak	No
C462	Medium	Irregular	Active	Weak	No
C463	Medium	Regular	Passive	Strong	No
C464	High	Irregular	Passive	Weak	Yes
C465	Medium	Irregular	Passive	Weak	Yes
C466	Low	Irregular	Passive	Strong	Yes
C467	High	Irregular	Active	Strong	No
C468	Low	Regular	Active	Weak	Yes
C469	High	Irregular	Passive	Weak	No
C470	Low	Regular	Active	Strong	No

Dataset C records 471–480

Table 75. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C471	High	Regular	Passive	Weak	No
C472	Medium	Regular	Passive	Weak	Yes
C473	High	Regular	Passive	Weak	No
C474	Medium	Regular	Passive	Weak	Yes
C475	High	Regular	Active	Strong	No
C476	High	Regular	Active	Strong	No
C477	High	Regular	Active	Weak	No
C478	Low	Irregular	Passive	Strong	Yes
C479	Medium	Regular	Active	Strong	No
C480	Medium	Regular	Passive	Weak	No

Dataset C records 481–490

Table 76. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C481	Low	Irregular	Active	Strong	Yes
C482	Low	Irregular	Passive	Weak	Yes
C483	Low	Regular	Passive	Strong	No
C484	Medium	Regular	Active	Strong	No
C485	Low	Regular	Passive	Strong	No
C486	Low	Regular	Passive	Weak	Yes
C487	Medium	Irregular	Active	Weak	No
C488	Medium	Irregular	Passive	Strong	No
C489	Low	Regular	Active	Strong	No
C490	Medium	Regular	Passive	Weak	Yes

Dataset C records 491–500

Table 77. Supporting data and analysis

ID	Attendance	Assignments	LMS	PriorCGPA	AtRisk
C491	Medium	Regular	Active	Strong	No
C492	Medium	Regular	Active	Weak	No
C493	Low	Regular	Passive	Strong	Yes
C494	Medium	Regular	Active	Strong	No
C495	Low	Irregular	Passive	Strong	No
C496	Low	Regular	Active	Strong	Yes
C497	High	Regular	Passive	Strong	Yes
C498	Low	Irregular	Active	Strong	No
C499	High	Irregular	Passive	Strong	Yes
C500	Medium	Regular	Active	Weak	No

APPENDIX D – GLOSSARY AND VALIDATION CHECKLIST

Table 78. Supporting data and analysis

Term	Meaning
Accuracy	Fraction of all predictions that are correct.
AtRisk positive class	The Yes label used as the positive class for recall and precision.
Candidate Elimination	An algorithm that maintains general and specific boundaries of consistent hypotheses.
Concept learning	Learning a Boolean target concept from labelled examples.
Entropy	A measure of class uncertainty in a set of examples.
False negative	An actually at-risk student predicted as not at risk.
False positive	A student predicted at risk who is not labelled at risk.
Gain ratio	Information gain divided by split information.
Hypothesis space	The set of candidate functions allowed by the representation.
Inductive bias	Assumptions that allow a learner to generalise beyond observed examples.
Information gain	Reduction in entropy obtained by splitting on an attribute.
Precision	Fraction of predicted positives that are truly positive.
Recall	Fraction of actual positives that are identified.
Version space	All hypotheses consistent with the observed training examples.

D.1 Final quality-control checklist

- ☐ All task headings match the assignment's Task 1–11 structure.
- ☐ Dataset A, B and T values are transcribed from the assignment handout.
- ☐ Dataset C is labelled synthetic and its generator and seed are documented.
- ☐ All tables have headers, consistent alignment and sequential numbering.
- ☐ All model metrics identify the positive class and the evaluation split.
- ☐ Unknown categories are handled explicitly rather than silently coerced.
- ☐ StudentID is excluded from predictive features.
- ☐ Any claimed 90% recall is supported by a sufficiently large independent validation set.
- ☐ All flagged decisions are accompanied by a human-readable path and review option.